The texdef script

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CTAN: http://www.ctan.org/pkg/texdef

VC: https://sourceforge.net/projects/texdef/

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Abstract

This Perl script shows the definition of (LA)TEX commands in a similar way as the TEX primitive `\show`. It is intended to allow users to quickly see the definitions of user level or internal package macros as well as the values of registers. The given commands are compiled internally with (LA)TEX and the output of `\show` is formatted and shown to the user.

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1 Usage

The script executable is called `texdef` (or maybe `texdef.pl`). A normal installation is supposed to also install it as `latexdef` (e.g. using a symlink to `texdef`) in order to allow for quick access to the LATEX format. Further format shortcuts are possible as explained in the installation section.

```
texdef [⟨options⟩] ⟨commandname⟩ [⟨commandname⟩ ...]
latexdef [⟨options⟩] ⟨commandname⟩ [⟨commandname⟩ ...]
```

Other program names are possible. See the `tex` option. Command names do not need to start with '\'.

Options

The following options can be used:

`-tex` ⟨*format*⟩, `-t` ⟨*format*⟩

Use given format of TeX: 'tex', 'latex', 'context'. Variations of 'tex' and 'latex', like 'luatex', 'lualatex', 'xetex', 'xelatex' are supported. The postfix '-dev' for develop versions of the format is also supported (e.g. 'latex-dev'). The default is given by the used program name: 'texdef' -> 'tex', 'latexdef' -> 'latex', etc.

`-texoptions` ⟨*options*⟩

Call LaTeX/TeX with the given options.

`-source, -s`

Try to show the original source code of the command definition (L).

`-value, -v`

Show value of command instead (i.e. `\the\command`).

`-preamble, -P`

Show definition of the command inside the preamble.

`-Environment, -E`

Every command name is taken as an environment name. This will show the definition of both `\foo` and `\endfoo` if `foo` is used as command name (L).

`-beforeclass, -B`

Show definition of the command before `\documentclass`.

`-package` ⟨*pkg*⟩`, -p` ⟨*pkg*⟩

(M) Load given tex-file, package or module depending on whether '*tex', '*latex' or 'context' is used. For LaTeX the ⟨*pkg*⟩ can start with '[<options>]' and end with ⟨*pkgname*⟩ or `{`⟨*pkgname*⟩`}`.

`-class` ⟨*class*⟩`, -c` ⟨*class*⟩

(LaTeX only) Load given class instead of default ('article'). The ⟨*class*⟩ can start with `[`⟨*class options*⟩`]` and end with ⟨*classname*⟩ or `{`⟨*classname*⟩`}`.

`-environment` ⟨*env*⟩`, -p` ⟨*env*⟩

(M) Show definition inside the given environment ⟨*env*⟩.

`-othercode` ⟨*code*⟩`, -o` ⟨*code*⟩

(M) Add other code into the preamble before the definition is shown. This can be used to e.g. load PGF/TikZ libraries.

`-before` ⟨*code*⟩`, -b` ⟨*code*⟩

(M) Place ⟨*code*⟩ before definition is shown. The ⟨*code*⟩ can be arbitray TeX code and doesn't need be be balanced.

`-after` ⟨*code*⟩, `-a` ⟨*code*⟩

(M) Place ⟨*code*⟩ after definition is shown. The ⟨*code*⟩ can be arbitray TeX code and doesn't need be be balanced.

`-find, -f`

Find file where the command sequence was defined (L).

`-Find, -F`

Show full filepath of the file where the command sequence was defined (L).

`-list, -l`

List user level command sequences of the given packages (L).

`-list-defs, -L`

List user level command sequences and their shorten definitions of the given packages (L).

`-list-all, -ll`

List all command sequences of the given packages (L).

`-list-defs-all, -LL`

List all command sequences and their shorten definitions of the given packages (L).

`-ignore-cmds` ⟨*cs,cs,..*⟩, `-i`

Ignore the following command sequence(s) in the above lists. (M)

`-ignore-regex` ⟨*regex,..*⟩, `-I`

Ignore all command sequences in the above lists which match the given Perl regular expression(s). (M)

`-pgf-keys, -k`

Takes commands as `pgfkeys` and displays their definitions. Keys must use the full path but the common '.cmd' prefix is applied.

`-pgf-Keys, -K`

Takes commands as `pgfkeys` and displays their definitions. Keys must use the full path.

`-version, -V`

If used alone prints version of this script. (L) Together with -p or -c prints version of LaTeX package(s) or class, respectively.

`-edit`

Opens the file holding the macro definition. Uses `--Find` and `--source`. If the source definition can not be found the definition is printed as normal instead. (L)

`-editor <editor>`

Can be used to set the used editor. If not used the environment variables `TEXDEF_EDITOR`, `EDITOR` and `SELECTED_EDITOR` are read in this order. If none of these are set a list of default editors are tried. The `<editor>` string can include '`%f`' for the filename, '`%n`' for the line number and '`%%`' for a literal '`%`'. If no '`%`' is used '`+%n %f`' is added to the given command.

`-tempdir <directory>`

Use given existing directory for temporary files.

`-help, -h`

Print this help and quit.

Notes:
Long option can be shorten as long the are still unique. Short options can be combined. If the option 'environment', 'before' and 'after' are used toegether the produced code will be inserted in the given order (reversed order for 'after').
(M) = This option can be given multiple times.
(L) = LaTeX only. Requires the packages 'filehook' and 'currfile'.

2 Installation

The latest official release of the script can be obtained from CTAN under http://mirrors.ctan.org/support/texdef/texdef.pl. The latest develop version (which can be unstable!) can be obtained from the source code repository under https://bitbucket.org/martin_scharrer/texdef/src/tip/texdef.pl.
For installation the script file `texdef.pl` must simply renamed to `texdef` and be copied to a location suitable for executables, i.e. a directory in the `PATH`, e.g. `/usr/bin` or `/usr/local/bin` for Linux/Unix. The script should be set as *executable*, i.e. `chmod +x texdef.pl` for Linux/Unix. This is not required for MS Windows.
This script can show the definitions of commands with different formats of TeX (`tex`, `etex`, `latex`, `pdftex`, `pdfetex`, `pdflatex`, `xetex`, `xelatex`, `context`, ...). The format can be given using an command line option or over the *script name*, i.e. if the script is called `texdef` it will use `tex`, but called `latexdef` it will use `latex` and so on. The script can be simply copied several times, but creating *symbolic links* to the real script is enough on operation systems/file systems which support this. At least the two variations `texdef` and `latexdef` should be installed.

Requirements

The `texdef` program is a Perl script which needs a Perl interpreter to work. Informations about installing Perl can be found at http://www.perl.org/get.html.
The script calls a TeX compiler (`tex`, `latex`, ...) to display the definition of commands/macros. Therefore (LA)TEX must also be installed. Informations about installing (LA)TEX can be found at http://www.latex-project.org/ftp.html.

Quick Install Nodes for Linux/Unix

```
cp texdef.pl /usr/local/bin/
cd /usr/local/bin
ln -s texdef.pl texdef
ln -s texdef.pl latexdef
ln -s texdef.pl etexdef
ln -s texdef.pl luatexdef
ln -s texdef.pl lualatexdef
ln -s texdef.pl pdftexdef
ln -s texdef.pl pdflatexdef
ln -s texdef.pl xetexdef
ln -s texdef.pl xelatexdef
ln -s texdef.pl 'some other TeX format of your choice'
```

or use the Makefile (defaults to `/usr/local/bin`)

```
make install INSTALLDIR=/your/path
```

3 Examples

Show the definition of `\chapter` with different classes (`article` (default), `book` and `scrbook`):

```
latexdef chapter
latexdef -c book chapter
latexdef -c scrbook chapter
```

Show value of `\textwidth` with different class options:

```
latexdef -c [a4paper]{book} -v paperwidth
latexdef -c [letter]{book}  -v paperwidth
```

Show definition of TikZ's `\draw` outside and inside a `tikzpicture` environment:

```
latexdef -p tikz draw
latexdef -p tikz --env tikzpicture draw
```

Show definition of TikZ's `\draw` inside a node, inside a `beamer` frame in handout mode:

```
latexdef -c [handout]beamer -p tikz --env frame \
    --env tikzpicture -b '\node {' -a '};' draw
```

List all user level command sequences (macros) defined by the `xspace` LaTeX package:

```
latexdef -l -p xspace
```

4 Changelog

v1.9 from 2020/09/27

- Added support for develop versions of the formats, e.g. `"latex-dev"`.

v1.8c from 2010/01/15

- Changed path printing format to native format of the used OS (e.g. under Windows file paths now habe '\' instead of '/')

v1.8a from 2018/03/28

- Further fixes of braces in regexs to avoid "Unescaped left brace in regex is deprecated" warnings or errors.

v1.8 from 2018/03/25

- Added `"dvitex"` and `"dvilatex"` as formats. These will set DVI mode on `"tex"` or `"latex"`.
- Escaped further braces in regexs to avoid warnings or errors.
- Added option `"--texoptions"` to allow the passing of compiler options for special cases.

v1.7c from 2017/12/09

- Fixed 'Unescaped left brace in regex is deprecated' warning. Thanks goes to Paulo Cereda <cereda.paulo@gmail.com> for providing the patch.

v1.7b from 2012/05/15

- Added support for `\chardef`, `\countdef`, `\dimendef`, `\mathchardef`, `\myskip`, `\skipdef` and `\toksdef`.
- Added support for `\newbox`, `\newcount`, `\newdimen`, `\newif`, `\newinsert`, `\newread`, `\newskip`, `\newtoks`, `\newwrite`.
- Added support for `\newif` and `\newcount`.
- Changed internal processing order to put more frequently used elements first.

v1.7a from 2012/05/08

- Fix for currfile v0.6.

v1.7 from 2012/05/07

- Changed "`--help`" to return status code 0.
- Changed Windows default editor to texworks. Fixed If-statement.
- Fixed bug which reported name of included subpackages.
- Added option "`--edit`".
- Added "`--editor`" option and better editor handling.
- Added variable substitution for editor.
- Added support for "`TEXDEF_EDITOR`" variable. Added Mac "`open`" command.
- Fixed for behaviour when -s is used (correct name in message; do not follow protected macros etc.)

v1.6 from 2012/05/02

- Changed implementation of -E option to work better with new environment source code.
- Added code to detect environment source definitions.
- Added -E option.
- Changed "`flavour`" to "`format`".
- Added support for `\let`.
- Added fallback for when the source code can not be found. Added support for -F with "`latex.ltx`".

v1.5 from 2012/04/29

- Added experimental –source option to show source code of the definitions.
- Script now works correctly under MS Windows.

v1.4 from 2011/07/28

- Added -F option to display file path.
- Added basic support for pdfkeys using -k and -K options.
- Added -V option to print version numbers of LaTeX packages or classes.
- Added '–tempdir' option to specify the location of the temporary files. Useful for debugging.